

1. Grade Checker

Q - Take a score as input and print the grade based on the following:

90+ : "A" , 80-89 : "B", 70-79 : "C", 60-69 : "D", Below 60 : "F"

here we used a basic if else statement to carry out marks and all.

Ans - Here users input the Marks then python code run and give output according to respective numbers, from Grade A,B,C,D or Fail.

```
Assignment > grade_checker.py > ...
1  grade = int(input('Enter your Marks: \n'))
2
3  if grade > 90:
4      result = 'Your Grade is A'
5  elif grade >= 80:
6      result = 'Your Grade is B'
7  elif grade >= 70:
8      result = 'Your Grade is C'
9  elif grade >= 60:
10     result = 'Your Grade is D'
11 else:
12     result = 'You are Fail'
13
14 print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 grade_checker.py
Enter your Marks:
89
Your Grade is B
○ sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$
```

2 Student Grades

Q - Create a dictionary where the keys are student names and the values are their grades. Allow the user to:

- (a) Add a new student and grade.
- (b) Update an existing student's grade.
- (c) Print all student grades.

Ans - (Q) Add a new student and grade. (Ans) Using (=) assignment operator , i simply assign a new key-value to the dictionary.

```
Assignment > student_grades.py > ...
1  student = {
2      'aman': 78,
3      'bablu': 56,
4      'chetan': 76,
5      'dharmender': 67
6  }
7
8  student['new_student'] = 49;
9  print(student);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 student_grades.py
{'aman': 78, 'bablu': 56, 'chetan': 76, 'dharmender': 67, 'new_student': 49}
○ sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$
```

(Q) Update an existing student's grade. (Ans) Using the existing (key) dictionary , i simply update the old value to new value.

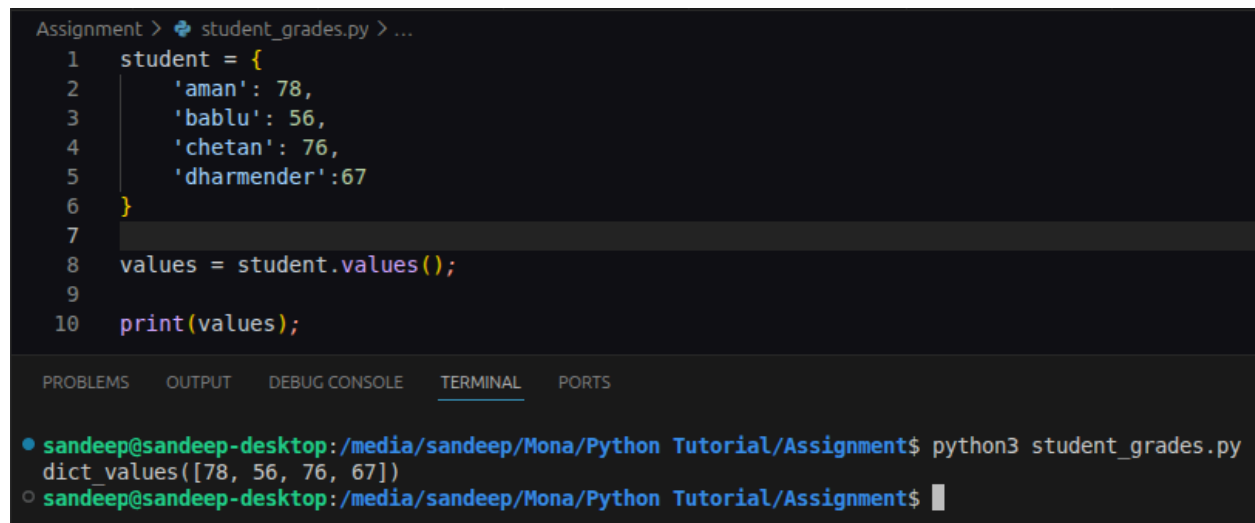
```
Assignment > student_grades.py > ...
1  student = {
2      'aman': 78,
3      'bablu': 56,
4      'chetan': 76,
5      'dharmender': 67
6  }
7
8  student['dharmender'] = 69;
9  print(student);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 student_grades.py
{'aman': 78, 'bablu': 56, 'chetan': 76, 'dharmender': 69}
○ sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$
```

(Q) Print all student grades.

(Ans) Using Python Dictionary values() Method, i simply print the student grades in the console.



The screenshot shows a code editor with a Python script named `student_grades.py`. The script defines a dictionary `student` with four entries: 'aman' (78), 'bablu' (56), 'chetan' (76), and 'dharmender' (67). It then uses the `values()` method to retrieve the grades and prints them. Below the code, the terminal output shows the command `python3 student_grades.py` being executed, resulting in the output `dict_values([78, 56, 76, 67])`.

```
Assignment > student_grades.py > ...
1 student = {
2     'aman': 78,
3     'bablu': 56,
4     'chetan': 76,
5     'dharmender': 67
6 }
7
8 values = student.values();
9
10 print(values);
```

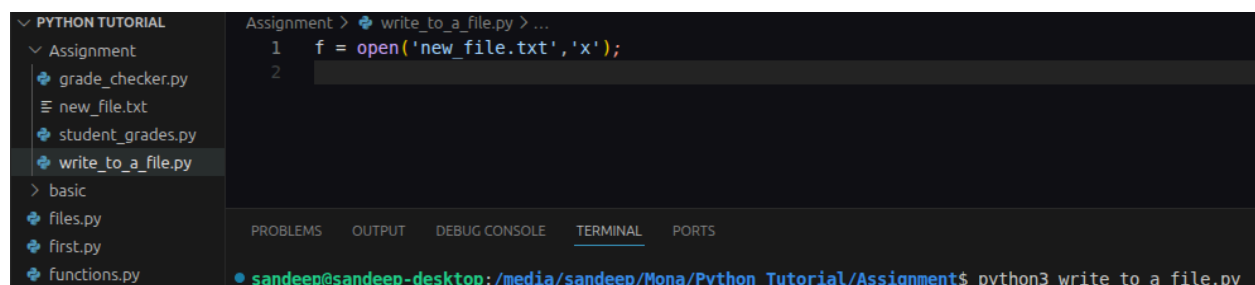
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
• sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 student_grades.py
dict_values([78, 56, 76, 67])
○ sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$
```

3. Write to a File

Q - Write a program to create a text file and write some content to it.

Ans - I use an open function to create a file, the first parameter is the file name, and second one is permission type (x) for creating the file, look inside the file is successfully created.



The screenshot shows a code editor with a Python script named `write_to_a_file.py`. The script uses the `open` function to create a new file named `new_file.txt` with 'x' permissions. The terminal output shows the command `python3 write_to_a_file.py` being executed successfully.

```
Assignment > write_to_a_file.py > ...
1 f = open('new_file.txt', 'x');
2
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
• sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 write_to_a_file.py
```

-> this s/shot tells about the code to write in (new_file.txt) file.

```
Assignment > write_to_a_file.py > ...
1 f = open('new_file.txt','w');
2
3 f.write('something is written in this file');
4 ✨
5 f.close

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 write_to_a_file.py
```

-> Look this s/shot the file is successfully written.

```
▼ PYTH... [🔍] [🔗] [🔄] [📄]
  ▼ Assignment
    ✨ grade_checker.py
    ✨ new_file.txt
    ✨ student_grades.py
    ✨ write_to_a_file.py
  > basic
    ✨ files.py
    ✨ first.py
    ✨ functions.py

Assignment > new_file.txt
1 something is written in this file

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 write_to_a_file.py
```

4. Read from a File

Q - We used open in read mode and file.read to read and print to display.

Ans - I use the read() function to read data from (new_file.txt), the data successfully printed in the console.

```
Assignment > write_to_a_file.py > ...
5 # f.write('something is written in this file');
6
7 f = open('new_file.txt');
8
9 print(f.read());
10
11 f.close

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$ python3 write_to_a_file.py
something is written in this file
○ sandeep@sandeep-desktop:/media/sandeep/Mona/Python Tutorial/Assignment$
```